

# Software Requirements Specification

Short Shelf Life

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# 1 Introduction

## 1.1 Purpose

This document describes the functional and non-functional requirements for the Short Shelf Life Application.

## 1.2 Project Description

The system allows users to create and manage books in their bookshelves, leave reviews, and view bookshelves of other users.

# 2 Functional Requirements

## 2.1 User Registration

**Description:** The system shall allow users to create an account using an email address and password only through valid email providers.

**Priority:** High

**Input:**

- Shelf Name
- Email address
- Password (salted hash)
- Re-enter Password

**Output:**

- Successful account creation message.
- Redirection to main bookshelf page.
- If the shelf has been populated, the books will be saved to the database instead of local storage.

**Validation Rules:**

- The email address must follow standard email format.
- Temporary or invalid email domains shall not be accepted.
- The system shall verify email uniqueness.
- The system shall protect against SQL injection attacks using parameterized queries.
- The system shall sanitize user-generated content to reduce XSS vulnerabilities.

## 2.2 User Login

**Description:** The system shall allow users to login with their verified email and passwords.

**Priority:** High

**Input:**

- Email address
- Password

**Output:**

- Successful login message
- Access to personal bookshelf
- Authentication tokens shall be securely stored in HTTP-only cookies.
- Authentication tokens will expire after a defined amount of time.

## 2.3 Add Books

**Description:** The system shall allow users to add a new book to their shelf. The default storage is the user's local storage.

**Priority:** High

**Input:**

- Book Name (0-30 characters, mandatory)
- Author Name (0-30 characters, mandatory)
- Review (0- 1300 characters, optional, default value empty)
- Color of book spine (from a color wheel, optional, default in place).
- Design of book spine (from an enum of designs, optional, default in place).

**Output:**

- Successfully added book message.
- Book will now be displayed on the shelf.
- Responsive design for long titles on the book spine.
- For longer titles, show the title with an ellipsis.

**Validation Rules:**

- The system shall protect against SQL injection attacks using parameterized queries.
- The system shall sanitize user-generated content to reduce XSS vulnerabilities.

## 2.4 View Books

**Description:** The system shall allow users to view the books they have added. If no review is added, a closed version will be shown. If a review is added, an open version will be shown.

**Priority:** High

## 2.5 Edit Books

**Description:** The system shall allow users to edit the title, author, and review of the book. The edit button will be placed on the Viewed Book, open and closed, depending on review status. Form constraints will still hold as outlined in the Add Book section.

**Priority:** High

**Input:**

- Edited Title (optional)
- Edited Author (optional)
- Edited Review (optional)

**Validation Rules:**

- The system shall protect against SQL injection attacks using parameterized queries.
- The system shall sanitize user-generated content to reduce XSS vulnerabilities.

## 2.6 Delete Book

**Description:** The system shall allow users to delete the book they have entered. Deletion would remove the book from local storage, database (wherever it is saved) and the shelf.

**Priority:** High

**Output:**

- Successful book deletion message

## 2.7 Browse Other Shelves

**Description:** The system shall allow users to view other shelves. Browsing will be done through a page where all the shelves and the name of shelves are displayed. Back button will take the user back to the browsing page then the user's own shelf. Users can interact with other shelves and read their reviews as well.

**Priority:** Medium

## 2.8 Interactive Effects

**Description:** The system shall allow users to interact with the main page in capacity outlined below.

**Priority:** Low

1. A toggle switch to turn light off and on.
2. Hovering over a book would highlight it.
3. Hovering over a book would show its full title.

## 2.9 Animation

**Description:** The system shall implement animation effects in some capacity, outlined below.

**Priority:** Low

1. Moving leaves.
2. Moving clouds during night.
3. Moving clouds during day.
4. Birds flying during day.

## 2.10 Background Music

**Description:** The system shall play a soft music in the background, with the choice to turn it off.

**Priority:** Low

**Priority:** Input

1. A toggle switch to turn music off and on.

## 2.11 System Time Background

**Description:** The system shall change the background according to the time in PKT. From 7pm to 6am the background would reflect night time. From 6.01 am to 6.59pm the background would reflect day time.

**Priority:** Low

## **3 Screens**

### **3.1 User Account Creation**

Would require Name, Email, Password.

### **3.2 Successful Account Creation**

After 3.1, depending on outcome.

### **3.3 Error During Account Creation**

After 3.1, depending on outcome.

### **3.4 User Login Page**

Would require Email, Password.

### **3.5 Error During Account Login**

After 3.4, depending on outcome.

### **3.6 Book Shelf Page**

1. Animation effects.
2. Interactive effects.
3. Button for music toggle.
4. Button for account creation.
5. Button for About.
6. Button to Add New Book.
7. Clicking on a Book would open the book. If no review, closed view. If review, open view.
8. View Other Shelves button.
9. Name of shelf visibly displayed.
10. Logout button
11. Edit Shelf Name and Password button.

### **3.7 Add Book**

Would require Title, Author, and ask for Review, Design.

### **3.8 View Book, Closed View**

Would have the title and the author. Would also have an Update and Delete Button.

### **3.9 View Book, Open View**

Would have the title, author and the review. Would also have an Update and Delete Button.

### **3.10 Successfully Added Book**

After 3.7, depending on the outcome.

### **3.11 Error While Adding Book**

After 3.7, depending on the outcome.

### **3.12 Edit Book**

Would allow users to edit the title, author and review. Would also have a back button.

### **3.13 Successfully Updated Book**

After 3.12, depending on the outcome.

### **3.14 Error While Updating Book**

After 3.12, depending on the outcome.

### **3.15 Back Button Confirmation Message**

If the user presses on the back button during editing.

### **3.16 Delete Book**

Would allow users to delete the book. Confirmation will be asked first.

### **3.17 Successfully Deleted Book**

After 3.16, depending on the outcome.

### **3.18 Error While Deleting Book**

After 3.16, depending on the outcome.

### **3.19 Deletion Confirmation Message**

Deletion will proceed only if the user presses Yes.



### **3.20 Other Shelves Page**

A consolidated page with other shelves modeled as books. Each shelf would show the user name. Clicking on the shelf would allow you to see the other shelf. Other shelf would have a back button to return to the browse page. Browse Page would allow you to return to your shelf.

### **3.21 Edit Profile Page**

Users can edit shelf name and password. Will also have a back button to cancel edits.

### **3.22 Confirmation of Changes**

Edits will only be saved if user clicks yes.

### **3.23 Back butto confirmation**

Edits will be discarded if user clicks yes.

### **3.24 Main Landing Page**

Will feature three options: Login, Register, Proceed as Guest.